



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                 | Progress Achieved                                                                          |
|----------------------------------|--------------------------------------------------------------------------------------------|
| Finalize project topic and scope | Selected “Vision-Based Smart Traffic Management System” using YOLO, Python, and Streamlit. |
|                                  | Defined modular architecture and real-time goals.                                          |
|                                  |                                                                                            |
|                                  |                                                                                            |
|                                  |                                                                                            |
|                                  |                                                                                            |
|                                  |                                                                                            |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                                     | Progress Achieved                                                                             |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Conduct literature review on existing traffic system | Reviewed traditional signal systems, inductive loop detectors, and vision-based alternatives. |
|                                                      | Identified gaps in adaptability and cost-efficiency.                                          |
|                                                      |                                                                                               |
|                                                      |                                                                                               |
|                                                      |                                                                                               |
|                                                      |                                                                                               |
|                                                      |                                                                                               |

Guides Review: \_\_\_\_\_  
\_\_\_\_\_

Signature Name and Signature of Team

Member 1: Name and Signature of Team

Member 2: Name and Signature of Team

Member 3: Name and Signature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned              | Progress Achieved                                                                                                     |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Select and trained YOLO model | Chose YOLOv8 for its speed and accuracy. Installed dependencies and verified model inference on sample traffic images |
|                               |                                                                                                                       |
|                               |                                                                                                                       |
|                               |                                                                                                                       |
|                               |                                                                                                                       |
|                               |                                                                                                                       |
|                               |                                                                                                                       |
|                               |                                                                                                                       |

Guides Review: \_\_\_\_\_  
\_\_\_\_\_

Signature Name and Signature of Team

Member 1: Name and Signature of Team

Member 2: Name and Signature of Team

Member 3: Name and Signature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                           | Progress Achieved                                                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Collect and organize traffic image dataset | Created folder-driven structure for multiple signals. Curated lane-wise images simulating real-world traffic conditions. |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |
|                                            |                                                                                                                          |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                 | Progress Achieved                   |
|----------------------------------|-------------------------------------|
| Build vehicle detection pipeline | Integrated YOLO with Python to      |
|                                  | classify traffic levels (Jam, High, |
|                                  | Medium, Low, Empty) and extract     |
|                                  | confidence scores.                  |
|                                  |                                     |
|                                  |                                     |
|                                  |                                     |
|                                  |                                     |
|                                  |                                     |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                | Progress Achieved                                                                                            |
|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| Design green light timing logic | Mapped traffic levels to base durations. Developed formula to adjust timing using confidence-weighted logic. |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |
|                                 |                                                                                                              |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                      | Progress Achieved                                                                                  |
|---------------------------------------|----------------------------------------------------------------------------------------------------|
| Develop Streamlit dashboard interface | Built interactive dashboard showing lane images, predictions, signal timers, and auto-cycle logic. |
|                                       | Enabled multi-signal support                                                                       |
|                                       |                                                                                                    |
|                                       |                                                                                                    |
|                                       |                                                                                                    |
|                                       |                                                                                                    |
|                                       |                                                                                                    |
|                                       |                                                                                                    |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                          | Progress Achieved                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Implement countdown and cycle advancement | Added per-signal countdowns with session state tracking. Automated batch switching after timer completion. |
|                                           |                                                                                                            |
|                                           |                                                                                                            |
|                                           |                                                                                                            |
|                                           |                                                                                                            |
|                                           |                                                                                                            |
|                                           |                                                                                                            |
|                                           |                                                                                                            |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                    | Progress Achieved                                                                                              |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Test system across multiple signals | Simulated multi-intersection behavior. Verified image loading, prediction accuracy, and timer synchronization. |
|                                     |                                                                                                                |
|                                     |                                                                                                                |
|                                     |                                                                                                                |
|                                     |                                                                                                                |
|                                     |                                                                                                                |
|                                     |                                                                                                                |
|                                     |                                                                                                                |

Guides Review: \_\_\_\_\_  
\_\_\_\_\_

Signature Name and Signature of Team

Member 1: Name and Signature of Team

Member 2: Name and Signature of Team

Member 3: Name and Signature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_



**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING  
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)**

**Academic Year: 2025 – 2026**

Class: \_\_\_\_\_

Sem: \_\_\_\_\_

Date: From \_\_\_\_\_ to \_\_\_\_\_

| Progress Planned                                    | Progress Achieved                                                                                       |
|-----------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| Document system architecture and<br>finalize report | Created diagrams, wrote Report,<br>and compiled objectives, problem<br>statement, and solution overview |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |
|                                                     |                                                                                                         |

GuidesReview: \_\_\_\_\_  
\_\_\_\_\_

Signature Name andSignature of Team

Member 1: Name andSignature of Team

Member 2: Name andSignature of Team

Member 3: Name andSignature of Team

Member 4:

Name of Project guide: \_\_\_\_\_

Signature of guide with Date: \_\_\_\_\_